

MUHAMMAD FARHAT MEHDI

Computer Engineering Student | Computer Vision • AI/ML • Robotics

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#) | [Upwork](#)

PROFESSIONAL SUMMARY

Computer Engineering student at UET Taxila with hands-on experience developing robotics, computer-vision, machine-learning, and automation systems. Built ROS2/Nav2 mobile-robot workflows, OpenCV-enabled inspection, Jetson Nano and ESP32 integrations, speech/phonetic search, recommendation systems, and edge-AI prototypes. Strong Python foundation with FastAPI, Docker, PostgreSQL, Git/Linux, and end-to-end software-hardware integration.

EDUCATION

Bachelor of Science in Computer Engineering | University of Engineering and Technology (UET), Taxila, Pakistan

2023 - 2027

EXPERIENCE

Robotics Intern | Mobile Robotics & Autonomous Systems

Current

- Develop and integrate mobile-robot systems using ROS2, Jetson Nano, LiDAR, GPS, encoders, and embedded motor control.
- Support autonomous navigation, computer-vision integration, sensor interfacing, and custom robot monitoring/control dashboards.
- Test and debug software-hardware interfaces across navigation, perception, telemetry, and control components.

Freelance Automation Developer | GoHighLevel (GHL), n8n, Zapier

Current

- Design and implement CRM and workflow automations using n8n, GoHighLevel, Zapier, webhooks, and Python.
- Build lead-processing and business-process workflows that reduce repetitive manual work and improve operational consistency.

Remote Intern - Generative AI | Arch Technologies

Completed

- Completed a remote internship focused on Generative AI concepts, tools, and practical applications.

SELECTED TECHNICAL PROJECTS

RoboOps AI - Autonomous Robot Operations Platform | ROS2, Nav2, Gazebo, OpenCV, FastAPI, React, PostgreSQL, n8n, Docker

- Built an end-to-end platform linking local AI mission planning and human approval to ROS2/Nav2 autonomous execution, live telemetry, visual inspection, automation, and PDF mission reporting.

UGV Mobile Robot & Custom Dashboard | Jetson Nano, LiDAR, GPS, Encoders, ESP32, ROS2

- Integrated sensing, navigation, embedded control, and a custom dashboard for mobile-robot monitoring and operation.

Phonetic Search + Whisper | Python, Whisper, Soundex/Double Soundex, TF-IDF, Fuzzy Matching

- Developed speech-enabled search combining transcription, phonetic matching, text similarity, and fuzzy retrieval for imperfect spoken or typed input.

Edge AI Sensor Classification | Edge Impulse, Logistic Regression, ESP32, Sensor Data

- Built an embedded machine-learning workflow for sensor-data classification and on-device inference.

TECHNICAL SKILLS

Computer Vision & Robotics: OpenCV, ROS2, Nav2, Gazebo, Jetson Nano, LiDAR, GPS, encoders, sensor integration

AI & Machine Learning: Python, scikit-learn, Whisper, NLP, recommendation systems, logistic regression, Edge Impulse

Software & Backend: FastAPI, React, TypeScript, PostgreSQL, Redis, Docker, Linux, Git/GitHub, REST APIs

Automation & CRM: n8n, GoHighLevel, Zapier, Apollo, webhooks, workflow automation

CERTIFICATIONS & LANGUAGES

Certifications: Python; MATLAB; Python Interaction with OS; Automating Tasks with Zapier; Business Analysis Fundamentals; Aspire Leader Program Alumni.

Languages: English, Urdu, Saraiki